

Patient Shortage in Histopathology: Does Full-Feature Redundancy Explain the Residual Breadth Benefit?

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Abstract

At a fixed patch budget, histopathology patch classifiers trained on more patients are more accurate. On TCGA-UT, adjusting patch counts for within-patient redundancy leaves a residual benefit of about three percentage points per doubling of patients, but that adjustment measures dependence along a single principal direction of the features. Because TCGA-UT patches are repeated crops of a few uniform tumour regions per slide, their dependence may extend across many directions. This experiment replaces the single-direction correlation with a full-feature intraclass correlation, the share of total feature variance that lies between patients, and recomputes effective support on exactly the same patient and patch samples. The stored predictions of the breadth–depth grid remain unchanged, so no classifier is retrained. If the residual breadth benefit shrinks to practical irrelevance without degrading the fit of effective support, the residual reflects underestimated redundancy; if it persists, additional patients contribute information beyond their effective observations.

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1 Motivation

Patches from one patient share acquisition conditions and much of their morphology, so they are worth fewer independent observations than their number suggests. *Effective support* expresses this by dividing the patch count by a design effect that grows with the *intra-class correlation*, the share of feature variance that lies between patients rather than within them. The breadth–depth comparison of Queisler (2026) tested whether this adjustment explains why patient count matters. It trained multinomial logistic regression on frozen Virchow2 features using allocations that vary the number of patients and the patches per patient independently. At 160 patches per class, twenty patients contributing eight patches each outperformed five patients contributing 32 patches each by 9.52 percentage points on TCGA-UT (95% interval: 8.96 to 10.07). Even at equal effective support, each doubling of patients still added 2.98 points (2.47 to 3.46).

That residual need not indicate missing patient coverage. The correlation was computed on a scalar score, the projection of each feature vector onto the leading principal direction fitted across all 30 cancer types. TCGA-UT patches are rotated and rescaled crops of a few pathologist-selected uniform tumour regions per slide (Komura et al., 2022), so patches of one patient share tissue in every feature direction, not only along a direction that need not align with the variation within any single class. The scalar correlation may therefore understate redundancy. The breadth–depth grid cannot rule this out: correlations about 1.56 times as large as estimated would absorb the residual and fit the nine allocations equally well (Queisler, 2026).

The question is therefore: *does a correlation that uses all feature directions account for the residual breadth benefit on TCGA-UT?* If redundancy was underestimated, effective support computed from the full features should absorb the residual. If the residual persists, patients contribute variation that no redundancy adjustment of this form captures.

2 Study design

Only the redundancy measure changes. The design inherits the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features, three fixed patient-disjoint train–validation–test splits, and the nine allocations of Queisler (2026): 5, 10, or 20 patients per class, each contributing 8, 16, or 32 patches. TCGA-UT participants are called patients throughout. The five training draws per allocation, the logistic-regression predictions on the natural test partitions, and hence every cell accuracy stay fixed, so any change in the analysis is attributable to the correlation alone. Table 1 contrasts the two measures.

	Single-direction correlation	Full-feature correlation
Input	Projection onto the leading principal direction	All 2,560 feature coordinates
Dependence captured	Between-patient differences along one direction	Between-patient differences in total feature variance

Table 1: The two redundancy measures differ only in which feature variation they use. Both are computed from the same sampled patients and patches, both apply the same correction for unequal patient sizes, and both are evaluated against the same cell accuracies.

3 Full-feature correlation

The full-feature correlation follows the covariance-trace form of multivariate intraclass correlation (Shou et al., 2013): variance is summed over all coordinates first, and the correlation is

then formed from these totals. It is applied here to patch embeddings with patients as clusters, and serves as a descriptive measure of redundancy rather than as a validated effective sample size for classifier learning.

For one class in one training split, let H patients contribute n_i feature vectors x_{ij} , with $N = \sum_i n_i$ patches in total. Let $\bar{x}_i$ denote the mean feature vector of patient i and $\bar{x}$ the mean over all patches. Four quantities define the estimator. The *between-patient mean square* B measures how far patient means lie from the overall mean, weighting each patient by its number of patches. The *within-patient mean square* W measures how far patches lie from their own patient’s mean. Because patients contribute unequal numbers of patches, the *effective patient size* $\tilde{m}$ replaces the common patch count of a balanced design. Together they give the correlation $\hat{\rho}_{\text{full}}$:

$$\begin{aligned} B &= \frac{\sum_i n_i \|\bar{x}_i - \bar{x}\|^2}{H - 1}, & W &= \frac{\sum_i \sum_j \|x_{ij} - \bar{x}_i\|^2}{N - H}, \\ \tilde{m} &= \frac{N - \sum_i n_i^2 / N}{H - 1}, & \hat{\rho}_{\text{full}} &= \frac{B - W}{B + (\tilde{m} - 1)W}. \end{aligned} \quad (1)$$

For scalar scores, the squared norms reduce to squared differences and Equation (1) becomes the one-way random-effects estimator with unequal-cluster correction used for the single-direction correlation (Kenny and La Voie, 1985). The two measures thus differ only in their input.

The estimator reads intuitively. If a patient’s patches were independent draws around the class mean, patient means would scatter only through sampling noise, B would be close to W , and the correlation would be close to zero. If a patient’s patches were near-copies of each other, W would be small relative to B , and the correlation would approach one. Because each squared norm adds squared differences over all 2,560 coordinates, B and W collect patient-specific shifts from every feature direction, so a patient whose patches share tissue in directions other than the leading principal direction still raises the correlation. Directions with larger variance carry more weight. Rotating the feature basis leaves the estimate unchanged, whereas rescaling individual coordinates does not; the features therefore keep their original scale, without whitening or selection of directions.

Both measures use the same sample: for every class and training split, at most 200 patients and at most 200 patches per patient, drawn from the full training partition as in Queisler (2026). The two correlations therefore differ only in how dependence is summarized, not in which material it is measured on. Estimates are clipped to $[0, 1]$ before use, as for the single-direction measure, and raw values are reported. An estimate is undefined when fewer than two patients or no within-patient degrees of freedom are available, or when the denominator is not positive; such estimates are neither set to zero nor omitted.

For each measure, the clipped correlations of a class are averaged over the three splits to give $\bar{\rho}_c$. With G patients per class each contributing m patches, effective support of class c is

$$N_{\text{eff},c}(G, m) = \frac{Gm}{1 + (m - 1)\bar{\rho}_c}, \quad (2)$$

and the cell predictor $N_{\text{eff}}(G, m)$ averages $N_{\text{eff},c}$ over classes. Larger correlations reduce effective support most at large depth, where many patches come from each patient, so a larger full-feature correlation lowers the deep cells more than the broad ones.

4 Evaluation

The analysis asks whether a better measure of redundancy leaves less for patient count to explain. It refits the support surface of Queisler (2026) once with each correlation measure and compares how much breadth benefit remains.

4.1 Residual breadth benefit

Accuracy is patient-macro balanced accuracy: patch recall averaged first within each test patient and true class, then equally across patients and classes. A cell accuracy $A(G, m)$, in percent, averages the three splits and five draws equally. Across the nine cells, the surface

$$A(G, m) = \alpha + \beta \log N_{\text{eff}}(G, m) + \gamma \log G, \quad b = \gamma \log 2, \quad (3)$$

separates two contributions. The term $\beta \log N_{\text{eff}}$ captures what effective support explains, and $\gamma \log G$ captures what patient count adds beyond it. The *residual breadth benefit* b is the resulting gain in percentage points per doubling of patients at fixed effective support. With the single-direction correlation, b is 2.98 points (Queisler, 2026). If that correlation understated redundancy, the full-feature measure should lower the effective support of deep cells, attribute more of the breadth benefit to $\beta \log N_{\text{eff}}$, and shrink b . The surface is a descriptive summary of the nine cells, not a matched-support intervention.

A smaller b alone is not sufficient, because a correlation can shift the breadth benefit away from γ while making effective support a worse description of the grid. The analysis therefore also fits effective support alone, $A = \alpha + \beta \log N_{\text{eff}}$, and compares its residual standard deviation, with seven residual degrees of freedom, between the two measures. The full-feature measure counts as fitting no worse when this value does not exceed that of the single-direction measure, which was 0.25 points (Queisler, 2026). Neither correlation is adjusted to improve the fit. Class correlations and cell effective support are reported alongside, showing where the two measures disagree.

4.2 Uncertainty

Two sources of uncertainty are reported separately, because they answer different questions.

Test-patient uncertainty asks how much the result depends on which patients happen to form the test partitions. Each of the 2,000 paired Bayesian bootstrap replicates of Queisler (2026) assigns Dirichlet weights to the distinct test patients (Rubin, 1981). A patient’s weight is shared across both measures, all cells, draws, splits, classes, and repeated test appearances, so differences between the measures are not blurred by different resamples. Accuracies and surfaces are recomputed with correlations held fixed, and the 2.5th and 97.5th percentiles give exploratory 95% intervals.

Training-patient sensitivity asks how much the result depends on which training patients entered the correlation estimates. A separate set of 2,000 cluster bootstrap replicates resamples the distinct patients in the correlation samples with replacement. A resampled patient keeps all of its sampled patches in every class and split in which it appears, and both measures use the same resample. Correlations, effective support, and surfaces are recomputed, while cell accuracies stay fixed. These intervals are not combined with the test-patient intervals and do not describe retraining on new cohorts. Between-draw accuracy dispersion from the breadth–depth grid accompanies both.

5 Interpretation criteria

Conclusions rest on the test-patient interval for the full-feature residual b and on the practical threshold of one percentage point per doubling of patients (Queisler, 2026). Training-patient sensitivity accompanies each conclusion; a conclusion that changes under it is reported as fragile.

Underestimated redundancy. If the interval for b lies within $[-1, 1]$ points and effective support alone fits no worse than with the single-direction correlation, the residual breadth benefit is attributable to redundancy that the leading principal direction missed. Full-feature effective support would then be the more complete summary of patient shortage on TCGA-UT.

Breadth beyond redundancy. If the interval lies above one point, additional patients retain a practically relevant benefit that this redundancy measure does not explain. Patients would then contribute variation beyond their effective observations, such as additional morphology or institution-specific acquisition; the present design does not identify which.

Inconclusive outcomes. All other outcomes, including an interval that crosses a threshold or a small residual accompanied by a worse fit of effective support alone, leave both accounts open.

Neither outcome establishes a causal mechanism. Both measures compress dependence into one scalar per class and assume that patches within a patient are exchangeable. All results are conditional on Virchow2 features, logistic regression, the sampled training patients, and the range of the nine allocations. Interventions on institutions, comparisons of representations, and methods that mitigate patient shortage lie outside the scope of this experiment.

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